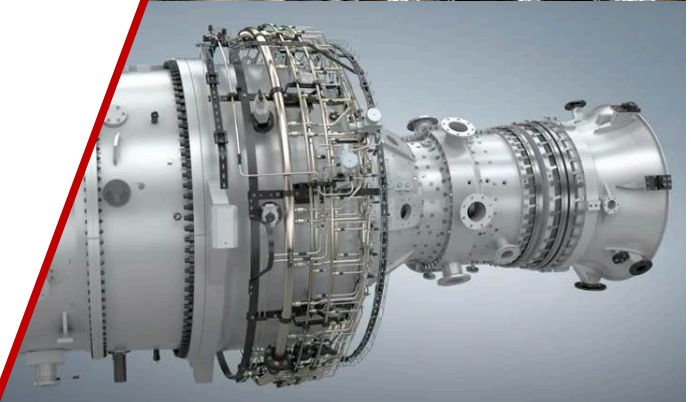
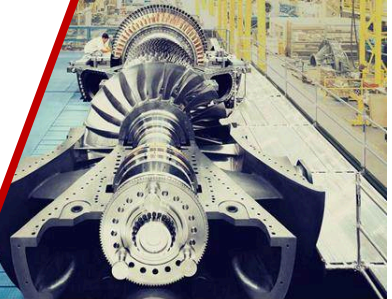




# Power Generation

Manufacturing Special Bolting  
and CNC Machined Components



SCAN ME

 Quality is our Hallmark

## The Cooper Turner Beck Group

Established over 150 years ago, we are manufacturers of special bolting and CNC machined components for use in **Safety Critical Applications**.

We manufacture bespoke components to recognised standards such as BS, DIN, ISO, ANSI, or to customer specific requirements by means of drawings or samples.

Our scope of supply covers all materials, and we specialise in components for use in Heat Resistant, Pressure Resistant or Corrosion Resistant applications.

**Our mission is simple:** to supply the very best quality products at competitive prices, always being flexible and agile, but more importantly being easy to do business with.

## Manufacturing....an enviable reputation

We are able to offer a wide range of production options to suit your unique requirements in terms of quantity, type of product and material.

**Quantity:** We manufacture to order, ranging from one-offs, to small and medium batch quantities.

**Size Range:** M2 to M200 (and imperial equivalents), all lengths available (lengths in excess of 4 metres can be catered).

**Hot Forging:** We can manufacture headed components (Hex Bolts, Socket Head Capscrews, etc.) in all grades of alloys.



## Service and Delivery...an unrivalled performance



We have a globally recognised proven track record in the supply of bespoke components for a diverse customer base.

Whatever the industry, whatever the demands, we are able to satisfy your requirements in terms of the level of service offered.

As a service orientated company, we strive to offer quality products and flexible lead times to our customers.

## Production...excellence in quality

- Thread forms:** All metric & imperial thread forms.  
**Bolts & Screws:** Hexagon, 12 Point, Socket, Slotted, Square.  
**Studs:** Double Ended, Waisted, Collar, Studbolts.  
**Nuts:** Hexagon, 12 Point, Capnuts, Round, Slotted, Barrel.  
**Washers:** Round, Square, Spherical, Spring, Taper, Tab.  
**Machined Parts:** Clevis Blocks & Pins, Tie Rods, Shafts, Spindles.



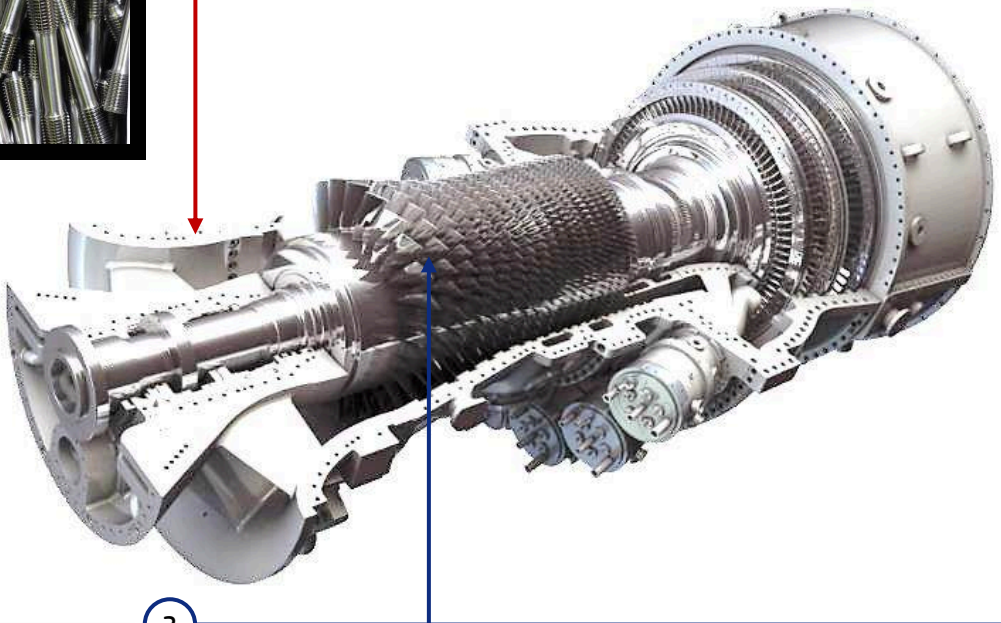
Safety critical fasteners are used in the rotors, combustion chamber, exhaust and casing of a gas turbine. They provide a critical role in the assembly of the turbine and they are highly specified and must be made to tight tolerances to ensure efficient and safe functioning



1

### Structural bolts for casings

- Fasteners are used to connect the main shaft to the impeller
- For casings around the blades, it requires high grade horizontal join bolts to hold the entire assembly in place
- Fasteners are also applied on the foundation of the gas turbine



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### Bespoke special bolting for use in rotor applications

- Rotor fasteners on rotor blades are used to attach to the rotating disk
- Specialty fasteners with highest criticality are used in combustor chamber due to very high temperatures
- Both specialty fasteners and standard bolts are used to fix compressor rows
- Fasteners bolt large sections together for exhaust applications





## Raw Materials...quality from the start

We have many years experience in the manufacture of bespoke components in a wide range of materials. Our range of strategic bar stock gives us the ability to offer a rapid delivery response time in order to meet customer requirements. Our scope of supply includes, but is not limited to, the grades listed below.

- |                   |                     |
|-------------------|---------------------|
| ▪ 304, 304L, 304H | ▪ Duplex 2205       |
| ▪ 316, 316L       | ▪ A193 B7, A193 L7  |
| ▪ 310             | ▪ Inconel 718 & 800 |
| ▪ 321             | ▪ Brass             |
| ▪ 410             | ▪ A36               |
| ▪ 904L            | ▪ 1008              |
| ▪ 317L            | ▪ 1018              |
| ▪ RA330           | ▪ 1020              |
| ▪ 347SST          | ▪ GR.50             |
| ▪ Alloy 20        | ▪ GR.55             |
| ▪ C276            | ▪ Copper            |
| ▪ Monel 400       | ▪ Aluminium         |



All manufactured components can be supplied with full material certification, to ensure full traceability of components throughout the production and certification processes.

Our safety critical fasteners meet the strict requirements and standards of the power generation industry. The following are important performance characteristics for fasteners for power generation:

- Excellent corrosion and oxidation resistance
- High tensile, shear, and fatigue strength
- Heavy-duty construction for high-strength applications
- High immunity to wear and vibration to prevent loosening
- High wear resistance
- Capable of operating under extremely harsh environments, including low and high temperatures and pressures

## Surface Coatings...the finishing touch

We are also able to supply products with additional processes such as surface coating and heat treatment, including but not limited to:

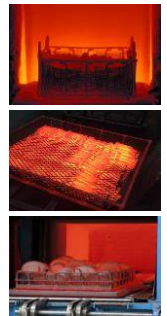
### SURFACE COATINGS

- Xylan / PTFE
- Zinc Plated
- HDG
- Phosphated
- Molykote
- Sherardized
- Silver Plated
- Nickel Plated
- Zinc Nickel Plated
- Copper Plated



### HEAT TREATMENT

- Case Hardening
- Tempering
- Plasma Nitriding
- Solution Annealing
- Age Hardening
- Nitro Carburising
- Normalising
- Strengthening
- Quenching



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